

SGA 2: Local Cohomology and Lefschetz Theorems

Exposé VIII: The Finiteness Theorem

Bounded source-aligned working unit — 19 July 2026

Authority. Corrected French lines 2901–2907; original printed p. 97; physical source-PDF pp. 84–85; recomposed running pp. 76–77. The unit crosses a recomposed page boundary, but the next **pageoriginale** marker is only at French line 2915, so its original printed-page coordinate remains p. 97. Blank line 2908 is excluded. French line 2909, the implication (a) $\Rightarrow$ (b), is the exact continuation cursor. The compiled French reader is generated from the same corrected edition and is page/layout evidence, not independent corroboration. The symbol Y is inherited from Corollary 2.3 and the Section 2 convention $U = X \setminus Y$; this contextual clarification is not a French-source emendation. Under closed Option A, target calligraphic $\mathcal{F}, \mathcal{G}$ are established English normalizations from source upright F, G , not literal glyph preservation.

Proof, opening.—Let $\mathcal{G}$ be a coherent $\mathcal{O}_X$ -module such that $\mathcal{G}|_U \simeq \mathcal{F}$ (EGA I, 9.4.3). Applying Corollary 2.3 to $\mathcal{G}$, we find that condition (a) is equivalent to (c):

(c) for every $x \in S$, one has

$$\mathrm{prof} \mathcal{F}_x > n - c(x), \quad c(x) = \mathrm{codim}(\overline{\{x\}} \cap Y, \overline{\{x\}}).$$